Epistemically Gated Weak-Form Discovery of Nonlinear PDEs Under Noise

Exact support recovery, null rejection, identifiability abstention, and cross-runtime reproducibility

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July 29, 2026 — revised September 3, 2026

All numbers in this note are generated programmatically from committed artifacts (frozen reference commit `8ecc3f85`, protocol v1.1); the build script aborts if any claim is not supported by the artifacts.

Abstract

We report a weak-form sparse-regression protocol for recovering the governing equations of one-dimensional nonlinear PDEs from field data, subjected to a strict epistemic discipline: every operator component is validated against analytical identities before use; the selection pipeline is calibrated on explicit null controls and then **frozen**; and every run is guarded against silent numerical-environment corruption. Under this frozen protocol (v1.1), the pipeline recovers the **exact governing support** of five nonlinear PDEs — Allen–Cahn, Fisher–KPP, Burgers, Korteweg–de Vries, and chaotic Kuramoto–Sivashinsky — under observational Gaussian noise from 0 to 10% of the field standard deviation, with maximum coefficient error 1.11% across all systems, noise levels and two independent seeds (7 and 11), while maintaining **zero false discoveries in 120 null-control runs** (temporal-shuffle and phase-randomized surrogates). A later sealed replica of the same ladder on 603 thermal-camera cases (423 opportunities to claim falsely) measured 14 false claims (3.3%), all sub-support claims on a term carrying about 0.5% of the dynamics and none on nulls; it motivated a declared resolution bound on every claim, confirmed on a second sealed panel (59 claims, 0 violations). The Kuramoto–Sivashinsky case is the primary evidence: the protocol, frozen before contact with the system, distinguishes the destabilizing anti-diffusion $-u_{xx}$ from the stabilizing hyper-diffusion $-u_{xxxx}$ on a spatiotemporally chaotic attractor. We further document a failure mode of stability selection — statistically stable supports on spectrum-preserving surrogates with relative fit residuals ≈ 0.80 — and its remedy, a residual-based epistemic gate. We do not claim discovery of new physics: the supported claim is that the protocol recovers known laws, rejects hard nulls, abstains when unidentifiable, and generalizes to a chaotic PDE never used to calibrate the pipeline.

1 Introduction

Sparse regression over libraries of candidate differential operators (SINDy [1] and successors, notably PDE-FIND [2]) can recover governing equations from data, and weak (integral) formulations [5, 6, 3, 4] extend this to noisy fields by shifting derivatives onto smooth test functions. Two chronic epistemic weaknesses remain. First, the numerical operator itself is usually trusted implicitly: discretization error in the weak features can masquerade as model mismatch and silently steer discovery. Second, model-selection stability is commonly treated as evidence of discovery, without null controls that measure how often the pipeline "discovers" structure where none exists.

This note documents a protocol that addresses both weaknesses by construction, and reports its behaviour on a five-PDE suite. The contributions are methodological and epistemic:

1. **Validate-before-discover.** No discovery is attempted until each weak operator component passes analytical identity gates on the target grid (Section 4), and the assembled operator passes an oracle ground-truth residual preflight ($r_{GT} < 10^{-3}$) on each dataset.

2. **Null-calibrated, then frozen.** The selection pipeline was calibrated once against null controls (which exposed a real failure mode, Section 7.1), amended to v1.1, and then frozen. Every subsequent system — including chaotic KS — was evaluated with zero recalibration.

3. **Runtime integrity as part of the evidence.** Each run embeds a machine-checked report proving the numerical environment was not affected by a silent array-mutation bug we discovered and quarantined (Appendix A).

2 Problem formulation

We observe a scalar field $u(x,t)$ on a periodic domain, sampled on a space–time grid, possibly with additive i.i.d. Gaussian measurement noise of standard deviation $\sigma \cdot \mathrm{std}(u)$. We posit a governing equation $u_t = \sum_j c_j\, \theta_j(u)$ with terms drawn from a candidate library. In weak form, for a compact test function ψ supported on a space–time window,

$$\int u_t\, \psi = -\int u\, \psi_t, \qquad \int \partial_x^m u\, \psi = (-1)^m \int u\, \partial_x^m \psi, \qquad \int u u_x\, \psi = -\tfrac{1}{2} \int u^2\, \psi_x,$$

so every feature is computed from field samples only — no derivatives of the data are ever estimated. The candidate library is shared across all systems, $\{u,\ u^2,\ u^3,\ u_x,\ u_{xx},\ u_{xxx},\ uu_x\}$, extended once (declared upfront, Section 5) with u_{xxxx} for the KS study. Each of $K = 200$ randomly placed windows contributes one row to the regression $Ac \approx b$.

3 Frozen discovery protocol (v1.1)

The protocol was preregistered, amended exactly once after null calibration (v1.0 $\to$ v1.1, Section 7.1, before any claim was formulated), and then frozen. For each dataset:

1. **Oracle preflight.** With the true coefficients c^*, compute $r_{GT} = \|Ac^* - b\|/\|b\|$. Discovery proceeds only if $r_{GT} < 10^{-3}$; otherwise the dataset resolution is insufficient for the operator and the run is refused. Resolution choices are made here, *before* selection ever runs.

2. **Stability selection** [9]**.** $B = 100$ subsamples of 60% of the windows; on each, STLSQ [1, 10] supports over a fixed threshold grid $\lambda \in \{0.02, 0.05, 0.1, 0.2, 0.4\}$ compete by BIC [11], **with the empty support always among the candidates** — the pipeline can abstain. A term enters the stable support if selected in $\geq 80\%$ of subsamples.

3. **Refit and residual gate.** Coefficients are refit on all windows restricted to the stable support. A discovery is *claimed* only if the support is stable **and** the relative fit residual $\|A_S c_S - b\|/\|b\|$ is below 5%: statistical stability alone is not treated as evidence (Section 7.1).

4. **Acceptance gates** (per system and noise level): exact support match, maximum relative coefficient error $< 5\%$, residual gate passed.

4 Numerical integrity and reproducibility

Componentwise identity gates. The temporal operator and each spatial identity (orders 1–3, plus the Burgers convective identity) were validated on analytical functions across quadratures, test functions and resolutions (9 temporal and 48 spatial configurations, all gates passed). The fourth-derivative identity required for KS was validated the same way before first use, including an analytical Faà di Bruno derivation of the bump test function's fourth derivative, verified against finite differences to 1.1×10^{-6} relative error.

Environment guard. During operator validation we discovered that on one interpreter stack (CPython 3.14.0 + NumPy 2.2.6, arm64), infix array operators inside functions can silently mutate their operand for arrays $\geq 256\,\text{KB}$ (temporary-elision defeated by interpreter stack references). This had produced a spurious "spatial-operator" failure signature in an earlier campaign. All runs in this note embed a runtime-guard report (canary PASS, supported interpreter matrix) in their evidence envelopes; the guard blocks execution otherwise, and did so once in production during this work (Appendix A).

Cross-runtime replication. The complete discovery campaign was re-run on an independent CPython 3.12.12 environment: **356/356 numeric fields bit-identical** to the 3.13.10 canonical run (operator validation campaign: 20/20 fields bit-identical).

Independent solver. To remove the same-generator objection, Burgers and Fisher–KPP fields were regenerated with a fully independent numerical chain (4th-order central finite differences + adaptive SciPy RK45 method-of-lines, versus the pseudo-spectral ETDRK4 [15] used elsewhere). Under the frozen protocol: exact support at every noise level, zero false discoveries in 20 nulls (preflight r_{GT}: 4.6×10^{-5} and 4.7×10^{-5}).

5 Experimental suite

System	Equation	Regime	Generator	r_{GT}
Allen–Cahn	$u_t = 0.05u_{xx} + u - u^3$	bistable	ETDRK4	1.5×10^{-4}
Fisher–KPP	$u_t = 0.05u_{xx} + u - u^2$	reaction–diffusion	ETDRK4	1.0×10^{-4}
Burgers	$u_t = -uu_x + 0.08u_{xx}$	nonlinear advection	ETDRK4 (+FD4/RK45)	4.6×10^{-5}
KdV [14]	$u_t = -uu_x - 0.0484\,u_{xxx}$	dispersive	complex-L ETDRK4	8.6×10^{-5}
Kuramoto–Sivashinsky [12, 13]	$u_t = -uu_x - u_{xx} - u_{xxxx}$	chaos, $L = 32\pi$	ETDRK4	8.7×10^{-6}

Noise ladder: $\sigma \in \{0, 1, 2, 5, 10\}\%$ of $\text{std}(u)$, added to samples before feature construction. Null controls per system: 5 temporal frame shuffles and 5 phase-randomized surrogates (identical space–time spectral magnitude, Hermitian random phases, hence exactly real fields with no PDE dynamics). KdV and KS were never used to tune any component: KdV entered after the operator was validated (transfer), and KS entered after the protocol was frozen.

6 Results

Headline. Across all five systems, all noise levels, and two independent seeds (7 and 11): exact support recovery, maximum coefficient error 1.11%, maximum claimed-fit residual 4.1×10^{-2}, and **0**

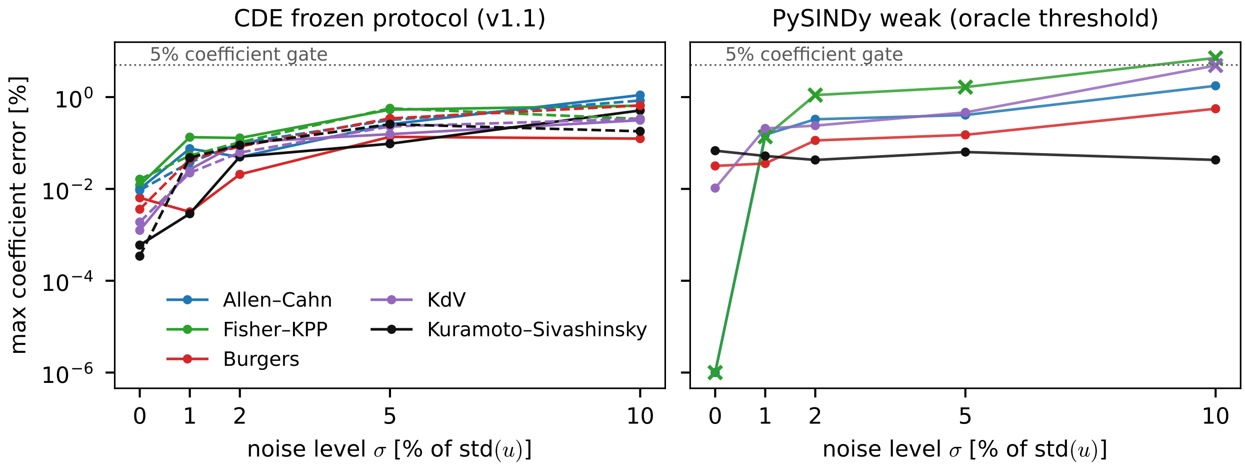

Figure 1: Maximum relative coefficient error versus noise level, from committed artifacts. Left: frozen CDE protocol (solid: seed 7; dashed: seed 11); all 25 (system, σ) cells are claimed and pass the 5% gate. Right: paired PySINDy weak-form baseline with oracle threshold selection; crosses mark cells whose recovered support is not exact.

false discoveries in 120 null runs. Figure 1 traces coefficient accuracy across the noise ladder, for the frozen protocol and the paired PySINDy baseline.

Seed characterization (20 fresh seeds). A preregistered multi-seed campaign — seeds 101–120, protocol frozen and committed before any seed was run, seeds 7 and 11 excluded from the estimate *because they produced the original claim* — gives, for the four V8 systems: **exact support recovery in 20/20 seeds at every noise level up to 10%**, and **0 false discoveries in 800 null runs**. Median maximum coefficient error at $\sigma = 10\%$ ranges from 1.9×10^{-3} (KdV) to 1.7×10^{-2} (Allen–Cahn). Support recovery is therefore substantially more robust than two seeds could establish.

The epistemic gate, however, is not. On Allen–Cahn at $\sigma = 10\%$ the gate rejects a *correct* recovery in **9 of 20 seeds**, although the support is exact in all 20. Seeds 7 and 11 both happen to pass, which is why an earlier version of this note stated "no breakdown noise level reached ($\sigma^* > 10\%$ everywhere)": that is a property of those two realizations, not of the protocol, and we withdraw it. The failure direction is the safe one — the gate declines to certify a right answer rather than certifying a wrong one — but a conservative false-negative rate of 45% at high noise on one system is a real cost, and it was invisible at two seeds.

The chaotic case, characterized separately, does not share this weakness. A second preregistered campaign put KS through the same twenty seeds under the same frozen criteria, with one criterion added in advance — gate survival at $\sigma = 10\%$ — specifically because the V8 campaign had just identified the gate as the fragile component. KS returns **exact support recovery in 20/20 seeds at every noise level**, **gate survival in 20/20 seeds at** $\sigma = 10\%$, σ^* reached in no seed, 0 false discoveries in 200 null runs, and a median maximum coefficient error at $\sigma = 10\%$ of 2.1×10^{-3}, an order of magnitude below Allen–Cahn's. The gate that declines 9 of 20 correct recoveries on the smooth relaxing field declines none on the broadband chaotic one. We had registered the opposite expectation before running. The result matters less as a prediction failure than as independent support, on forty seeds that did not exist when the observation was first made, for the asymmetry of §7.2: residual gating strains precisely where stability selection is least discriminative on its own.

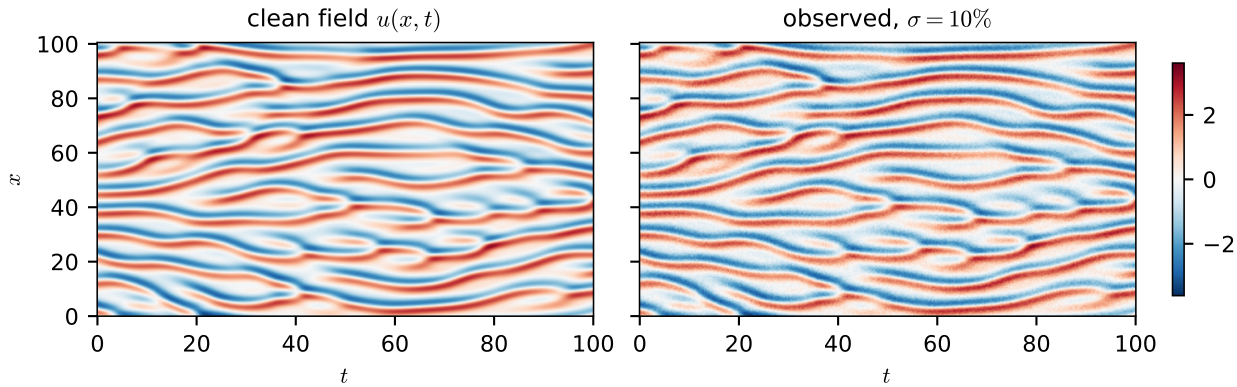

Figure 2: The chaotic Kuramoto–Sivashinsky field behind the primary evidence (seed 7): clean field $u(x,t)$ and the $\sigma = 10\%$ observation seen by the pipeline. The field is regenerated deterministically from the committed generator and verified against the committed grid metadata (N_x, N_t, L, $\mathrm{std}(u)$) before rendering.

6.1 Kuramoto–Sivashinsky (primary evidence)

Chaotic regime on $L = 32\pi$, transient discarded, 100 time units kept (Figure 2). The frozen pipeline recovers

$$u_t = -1.0000\, uu_x - 1.0000\, u_{xx} - 1.0000\, u_{xxxx} \qquad (\sigma = 0),$$

i.e. the **negative** (destabilizing) u_{xx} and the **negative** (stabilizing) u_{xxxx} jointly and with correct signs, out of an 8-term library on a chaotic attractor. At $\sigma = 10\%$ the coefficients remain $(-0.9992,\ -0.9981,\ -0.9949)$. This sign-structure discrimination under chaos is the strongest single piece of evidence in the suite — considerably stronger than a good fit on a smooth relaxing field.

σ	Support	Coefficients	Max err	Residual	Gate
0%	u_{xx}, u_{xxxx}, uu_x	u_{xx}: -1.0000, u_{xxxx}: -1.0000, uu_x: -1.0000	0.00%	7.9×10^{-6}	PASS
1%	u_{xx}, u_{xxxx}, uu_x	u_{xx}: -1.0000, u_{xxxx}: -1.0000, uu_x: -1.0000	0.00%	1.3×10^{-3}	PASS
2%	u_{xx}, u_{xxxx}, uu_x	u_{xx}: -1.0000, u_{xxxx}: -1.0005, uu_x: -0.9999	0.05%	2.6×10^{-3}	PASS
5%	u_{xx}, u_{xxxx}, uu_x	u_{xx}: -0.9991, u_{xxxx}: -0.9990, uu_x: -0.9992	0.10%	6.4×10^{-3}	PASS
10%	u_{xx}, u_{xxxx}, uu_x	u_{xx}: -0.9981, u_{xxxx}: -0.9949, uu_x: -0.9992	0.51%	1.3×10^{-2}	PASS

6.2 The four V8 systems

All four systems pass every gate at every noise level; coefficient tables in Appendix C. Summary at $\sigma = 10\%$ (worst noise):

System	Support (exact)	Max coef err	Fit residual
Allen–Cahn	u, u^3, u_{xx}	1.11%	4.1×10^{-2}
Fisher–KPP	u, u^2, u_{xx}	0.65%	5.3×10^{-3}
Burgers	u_{xx}, uu_x	0.12%	1.5×10^{-2}
KdV	u_{xxx}, uu_x	0.31%	2.6×10^{-2}

6.3 Null controls

120 null runs (frame-shuffled and phase-surrogate fields, all systems, both seeds where applicable, plus the external-solver arm): **zero claimed discoveries**. The decisive mechanism differs by null type and by field class (Section 7).

6.4 Paired external baseline (PySINDy weak-SINDy)

Setup (full disclosure). PySINDy [7, 8] v2.1.0: `CustomLibrary` $\{u, u^2, u^3\}$ wrapped in `WeakPDELibrary` (derivative order 3; 4 for KS), $K = 200$ weak subdomains, no bias term, PySINDy's STLSQ with ridge $\alpha = 10^{-5}$, threshold grid $\{0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$ with **oracle selection** per (system, noise level) — the best threshold judged against ground truth, an advantage the frozen CDE protocol never receives. Feature names mapped (`x0` $= u$); the library is PySINDy's natural weak set including all cross terms $f(u)\,\partial_x^k u$, a superset of ours. *Exact support* = all true terms nonzero and every other term zero. On nulls, the $\sigma = 0$-selected threshold is used (fixed policy).

Recovery. The baseline achieves exact support in 19/25 (system, σ) cells versus 25/25 for the frozen CDE protocol (failures: Fisher–KPP at every noise level, KdV at $\sigma \geq 5\%$). Where it recovers (Burgers, KS, Allen–Cahn), its oracle-tuned coefficients are accurate.

Nulls, and the causal decomposition. Vanilla PySINDy commits a false discovery on **49/50 null fields** (CDE: 0/120), even though an empty model was *reachable* in its threshold grid on 33/50 of them: STLSQ has no criterion that ever prefers it. We then ran the ablation **PySINDy + epistemic gate**, transplanting the v1.1 residual gate (proxy: $\sqrt{1-R^2}$ of PySINDy's own weak regression, same 5% level): null FDR drops to **0/50** — the null-rejection advantage is governance, and it transfers. But the same gate also rejects the baseline's *true* fits at moderate noise (only 14/25 cells remain claimed discoveries; KdV falls at $\sigma \geq 2\%$, Allen–Cahn and Fisher–KPP at $\sigma \geq 5\%$), while *all* 25 CDE cells pass the identical gate. The decomposition is clean: null rejection comes from the epistemic layer; noise-robust discovery power comes from the quality of the validated weak operator. Both are needed, and neither alone explains the paired result (Figure 3b).

7 Ablations and epistemic observations

7.1 Stability selection without the residual gate (v1.0 to v1.1)

The archived v1.0 run (no residual gate) shows the failure mode directly: phase-randomized surrogates produced *statistically stable* supports with per-system pre-gate false-discovery rates of 0.0 (Allen–Cahn), 0.3 (Fisher–KPP), 0.4 (Burgers), 0.5 (KdV). Chance correlations at dataset level survive window subsampling: stability is not evidence. Those stable-but-false fits, however, explain the data poorly: across all v1.1 campaigns the *minimum* fit residual of any stable null support is 0.80, versus 4.1×10^{-2} *maximum* for true systems — a separation of roughly four orders of magnitude (Figure 3a). The 5% residual gate, fixed once from this calibration and never touched again, reduces null FDR to 0 everywhere while leaving every true discovery untouched.

7.2 Smooth versus chaotic fields (empirical observation)

Pre-gate stable-support rates on nulls differ sharply by field class: smooth or quasi-stationary fields yield 0.30–0.50 (Fisher–KPP/Burgers/KdV; external-solver arm: 0.30–0.40), while the chaotic KS field yields 0.00 — on broadband chaos, the nulls do not even produce stable supports. **As an**

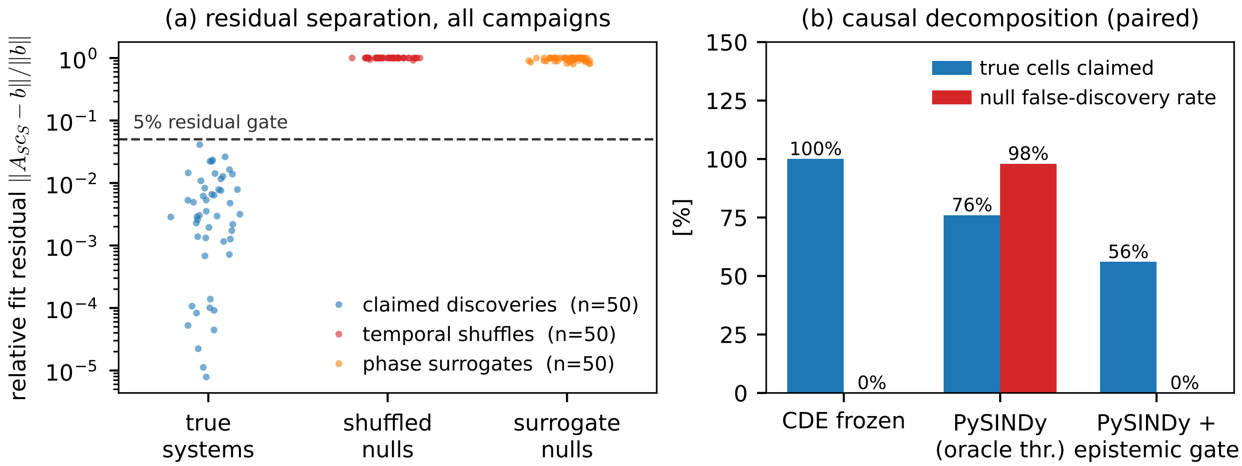

Figure 3: (a) Relative fit residual of every claimed discovery versus every null-control run across the v1.1 campaigns (counts in legend); the fixed 5% residual gate separates the two populations by roughly four orders of magnitude. (b) Paired causal decomposition: fraction of true (system, σ) cells claimed, and null false-discovery rate, for the frozen CDE protocol, the oracle-thresholded PySINDy baseline, and the same baseline with the transplanted epistemic gate.

empirical observation of this suite (not a universal principle): stability selection is intrinsically more discriminative on broadband chaotic fields than on smooth or quasi-stationary fields; residual gating is essential primarily in low-excitation regimes.

7.3 Oracle/identity preflight as a resolution oracle

The preflight caught two would-be failures before any discovery ran: KdV at the default resolution ($r_{GT} \approx 10^{-2}$, third-derivative quadrature-limited; opens at $N_x = 1024$), and KS at V8-standard sampling ($r_{GT} \approx 8 \times 10^{-2}$; ψ'''' on a chaotic field requires $N_x = 2048$, frame step 0.1, windows 12×7.5, giving $r_{GT} = 8.7 \times 10^{-6}$). Both decisions were made from the oracle residual alone, before selection — resolution tuning never saw the discovery output.

7.4 Empty-support abstention

Including the empty support among BIC candidates is what allows abstention: on pure-noise regressors the pipeline returns the empty model (committed unit test), and on KS nulls it abstains in 20/20 runs pre-gate. Without this, the threshold grid can never produce an empty candidate and the pipeline is structurally incapable of saying "nothing here" — a defect found by the null tests themselves.

8 Large-scale sealed replica and the declared resolution bound

Everything above concerns synthetic PDE fields and explicit null controls. Between the first version of this note and this one, the same discovery ladder (v2.1: non-oracle preflight, stability selection, residual gate, holdout transfer, relative swap test) was run on a much larger sealed panel of a different kind, and the result changes what the note may claim about false discoveries.

8.1 The replica: 603 sealed cases, 423 opportunities to claim falsely

The panel (preregistered 2026-08-02, generated with new seeds recorded in the ledger before generation) holds 603 thermal-camera-degraded diffusion fields in nine balanced classes: 268 representable (true support inside the library), 268 non-representable (a term outside the library), and 67 nulls. Two nodes ran it blind — macOS arm64 and Windows AMD64 with aligned NumPy/SciPy — with 100 cases in common. An *opportunity to claim falsely* is a null, a non-representable case, or a representable case on which the ladder produced a wrong support; the panel offered 423 of them, against a preregistered minimum of 350. The verdicts were produced on 2026-08-04 and scored on 2026-09-03 with the scorer of the earlier 90-case challenge (a false claim is a `CLAIM` on a null, on a non-representable case, or with a support different from the truth), eight mutation tests biting before the verdict was read.

Result: 14 false claims in 423 opportunities (3.3%; the rule-of-three bound the campaign hoped for was 0.71%). Zero on the 67 nulls, zero on the 268 non-representable cases; 105 correct claims on 268 representable cases (recall 39%, median coefficient error 1.17%). The two nodes agree on all 100 shared cases in verdict, support and blocking stage. The preregistered outcome is `CLAIM_FALSA_OSSERVATA`, which the preregistration itself called the campaign's purpose: "a campaign that can only confirm what it hopes for is not a test."

One failure class. All 14 false claims are in one class: nonlinear diffusion $D = \alpha(1 + \gamma v)$ with linear reaction, true support $\{v_{xx}, v, \partial_x(vv_x)\}$, claimed support $\{v_{xx}, v\}$. The missing term carries $\alpha\gamma \approx 5 \times 10^{-7}$ against $\alpha \approx 10^{-4}$: about half a percent of the dynamics. The two-term model fits within 1%, far below the 5% gate; the relative swap test (factor 2) cannot see a term at that level and, in 48 of the 67 cases of the class, correctly declares `NOT_IDENTIFIABLE`; in 14 it does not see the problem at all. We call this a *sub-support claim*: an effective model, exact up to a term below the gates' resolution, asserted as a law. It is the mirror image of the out-of-library Taylor surrogate: there a term outside the library masqueraded as two inside; here a term inside the library is too small to be seen.

What this changes in this note. The statements "zero false discoveries" in the abstract, Results and Conclusion refer to the as stated. They must not be read as a property of the ladder: on a sealed panel with representable cases at the resolution limit, the ladder's false-claim rate is 3.3%, concentrated entirely in sub-support claims. The seal of this replica is procedural, not cryptographic — the generation envelope did not record the truth hash and truth and verdicts sit in one commit; blindness rests on the discoverer's source, which never reads the sealed file, and on the recorded generation/execution order. Both weaknesses are fixed in the panel of the next subsection.

8.2 A declared resolution bound on every claim

The sub-support class asks for a statement the ladder was not making: how small a term the data could have hidden. For a claimed support S with fitted coefficients $\hat{c}$, design matrix A and time column b, we attach to every assertive verdict

$$X = k\,\frac{\|A_S\hat{c} - b\|_2}{\|b\|_2}, \qquad c_{\min}(t) = k\,\frac{\|A_S\hat{c} - b\|_2}{\|A_t\|_2} \;\; (t \notin S), \tag{1}$$

read as "law S up to library terms whose relative contribution $f_t = |c_t|\,\|A_t\|_2/\|b\|_2$ is below X". The bound changes no verdict; it changes what a verdict asserts. Soundness is a falsifiable property: for every claim on a representable case, every missing true term must satisfy $f_t < X$.

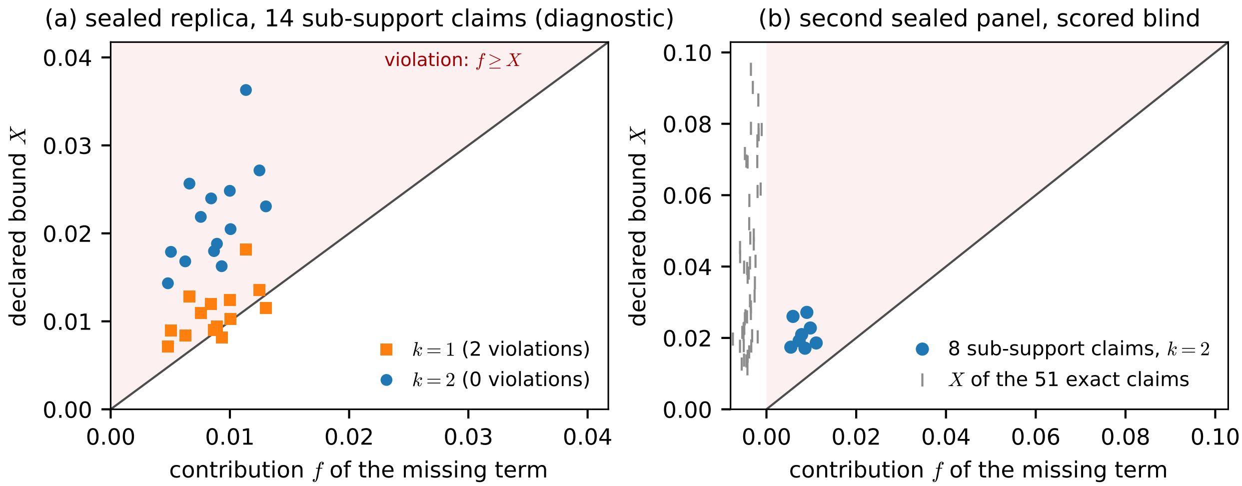

Figure 4: The declared resolution bound against the term it must cover. (a) The 14 sub-support false claims of the sealed replica: contribution f of the missing term $\partial_x(vv_x)$ against the bound X for $k = 1$ and $k = 2$; points above the diagonal violate soundness ($k = 1$: 2 violations; $k = 2$: none). (b) The 8 sub-support claims of the second sealed panel, scored blind with $k = 2$: f against X, all below the diagonal (margins 1.68–4.40×), with the X of the 51 exact claims as a marginal strip.

Choice of k (diagnostic, truth open). On the 14 false claims of the replica plus 30 of its correct claims, $k = 1$ is violated in 2 cases — the fitted model absorbs part of the missing term through correlated columns, so the residual understates it — while $k = 2$ (the identifiability factor already frozen in the ladder) is violated in none. The preregistration had predicted exactly this and fixed $k = 2$ before the confirmatory run; the diagnostic on the replica chose k, and confirmed nothing.

Confirmation on a second sealed panel. A new panel of 120 representable cases (four classes, new seeds), its truth hashed into the generation envelope and committed *before* the discoverer ran, was solved blind by the unchanged v2.1 ladder; predictions were hashed before the truth was opened; the scorer's mutation tests bit first. The ladder produced 59 claims: 51 exact and 8 strict false claims, all of them sub-support claims of the same class. **In all 8, the missing term sits below X:** margins from 1.68× to 4.40×, and the declared smallest detectable coefficient of the missing term (median 1.4×10^{-6}) exceeds its true value (median 4.8×10^{-7}) — the bound says, correctly, that the term was not resolvable. No verdict changed; median X is 3.0%. Verdict `CONFIRMED` by the preregistered rule (Figure 4).

What the bound does not do. It covers terms inside the library only: an out-of-library surrogate has no f_t and remains the business of the amplitude-extrapolation gate. The 8 strict false claims stay false by the exact-support definition, which we keep and report unchanged; the bound turns them from unqualified assertions into claims with a stated scope. Zero violations in 59 claims bounds the violation rate at 5.1% (rule of three): one sealed replication, not a validated property. The bound is now mandatory on every assertive verdict of the production pipeline (semantics amendment of 2026-09-03), and its soundness on the PDE pipeline of this note — identical formula, different features — is not yet measured.

9 Limitations

- **Synthetic 1D periodic data.** All fields are generated by known PDEs on periodic 1D domains; noise is i.i.d. Gaussian on samples. There are no real boundary conditions, no latent variables, no asynchronous or missing measurements, and no model misspecification beyond the library: conditions far cleaner than any sensor dataset. The protocol is validated within this perimeter, not yet on experimental science.

- **Library containment.** The true support is always contained in the candidate library; the protocol detects absence of structure (nulls) but was not tested against structured *wrong* libraries. The sealed replica of Section 8 shows the complementary failure: a term *inside* the library, too small to resolve, produces a sub-support claim; the declared resolution bound now scopes every claim to library terms above X, and says nothing about terms outside it.

- **Seeds — closed for the five headline systems.** The headline runs use two independent seeds (7 and 11). Two preregistered 20-seed campaigns (Results) now characterize recovery probability, gate survival and coefficient variance for all five systems — the four V8 systems and KS — and in doing so retracted one claim about the gate. An earlier version of this note attributed the gap on KS to its cost per seed; that was not measured and is false, KS being roughly five times cheaper per seed than the four-system run. The V10 line remains uncharacterized, and 40 seeds bound recovery probability only to within the resolution 40 seeds afford.

- **Oracle preflight requires the true coefficients**, so it is a *validation* instrument, not directly transferable to genuine discovery. The sealed replicas of Section 8 use its non-oracle replacement (quadrature agreement, test-function cross-consistency, window holdout); a process audit found its three thresholds set by hand at a flat 0.05 rather than from the calibration that had been preregistered for them, and that the quadrature test confounds measurement noise with discretization error above 10% noise. Both are recorded, neither is silently fixed. The identity gates remain applicable as-is.

- The smooth-vs-chaotic observation is from five systems; it is reported as an observation, not a law.

10 Conclusion

Under a frozen, null-calibrated protocol with componentwise-validated weak operators and runtime-integrity guards, exact support recovery of five nonlinear PDEs — including chaotic KS with opposing-sign diffusion terms — is achievable from noisy field data up to 10% noise with zero false discoveries in 120 null controls. That last figure is a property of the null controls, not of the ladder: a sealed replica on 603 cases measured a 3.3% false-claim rate, entirely from sub-support claims on a term below the gates' resolution, and the remedy — a declared resolution bound on every claim — held with zero violations on a second sealed panel. The epistemic machinery (identity gates, oracle preflight, abstention, residual gating, environment canaries, cross-runtime replication) is not overhead: each component caught at least one real failure during this program. The protocol does not discover new physics; it establishes a discipline under which claimed discoveries are hard to fake — including by accident.

A Environment contamination case

While debugging a persistent Burgers "spatial-operator" anomaly, we found that on CPython 3.14.0 + NumPy 2.2.6 (macOS arm64) infix array operations on ≥ 256 KB arrays inside functions mutate their operands in place (temporary elision defeated by interpreter stack references): `Uw ** 2` turned `Uw` into its square, so downstream features consumed U^4, U^6. The anomaly disappeared entirely (ground-truth residual $0.80 \to 4.6 \times 10^{-5}$) once the operations were rewritten via explicit ufuncs and the environment quarantined. A repository audit (4660 Python files; 809 flagged for review, 2 confirmed-and-fixed) and a mandatory runtime canary followed. During this very study, the guard blocked one run after a third-party install silently downgraded NumPy — the first production intervention of the guard. Full ledger and audit are committed alongside the campaigns.

B Reproducibility manifest

Run	Seed	Python	NumPy	Guard	`results.json` sha256
V7 (py3.13)	7	3.13.10	2.5.1	PASS	`a6f14e9ea5cabaf6`
V8 (py3.13)	7	3.13.10	2.5.1	PASS	`e5662ad857671b4d`
V8 (py3.12)	7	3.12.12	2.5.1	PASS	`e1704f0b5b24e45e`
External solver	7	3.13.10	2.5.1	PASS	`26439b27085048b5`
V9 KS (py3.13)	7	3.13.10	2.5.1	PASS	`9b4a119d6ed677d2`
V8 (seed 11)	11	3.13.10	2.5.1	PASS	`b359f86811322338`
V9 KS (seed 11)	11	3.13.10	2.5.1	PASS	`ed42bcc6dd6c4588`
V13 sealed replica (scored)	700000+	3.13.10	2.5.1	PASS	`04d08889eddb7fa2`
Blind-4 resolution bound	720000+	3.13.10	2.5.1	PASS	`945dc846e72f94dd`

Frozen reference commit: `8ecc3f85` (protocol v1.1). This note and its figures are generated by the committed build scripts (`build_research_note_latex_v0.py` and `build_note_figures_v0.py`), which re-read every artifact and assert every quantitative claim at build time.

C Extended coefficient tables

Allen–Cahn

σ	Support	Coefficients	Max err	Residual	Gate
0%	u, u^3, u_{xx}	u: +1.0000, u^3: −1.0000, u_{xx}: +0.0500	0.01%	1.4×10^{-4}	PASS
1%	u, u^3, u_{xx}	u: +1.0002, u^3: −0.9999, u_{xx}: +0.0500	0.08%	5.3×10^{-3}	PASS
2%	u, u^3, u_{xx}	u: +0.9999, u^3: −0.9995, u_{xx}: +0.0500	0.05%	7.8×10^{-3}	PASS
5%	u, u^3, u_{xx}	u: +1.0026, u^3: −0.9996, u_{xx}: +0.0500	0.26%	2.2×10^{-2}	PASS
10%	u, u^3, u_{xx}	u: +1.0111, u^3: −0.9998, u_{xx}: +0.0503	1.11%	4.1×10^{-2}	PASS

Fisher–KPP

σ	Support	Coefficients	Max err	Residual	Gate
0%	u, u^2, u_{xx}	u: $+1.0000$, u^2: -1.0000, u_{xx}: $+0.0500$	0.01%	9.2×10^{-5}	PASS
1%	u, u^2, u_{xx}	u: $+0.9999$, u^2: -0.9999, u_{xx}: $+0.0501$	0.13%	7.2×10^{-4}	PASS
2%	u, u^2, u_{xx}	u: $+1.0003$, u^2: -1.0003, u_{xx}: $+0.0499$	0.13%	1.2×10^{-3}	PASS
5%	u, u^2, u_{xx}	u: $+1.0000$, u^2: -0.9999, u_{xx}: $+0.0503$	0.53%	2.9×10^{-3}	PASS
10%	u, u^2, u_{xx}	u: $+1.0035$, u^2: -1.0039, u_{xx}: $+0.0503$	0.65%	5.3×10^{-3}	PASS

Burgers

σ	Support	Coefficients	Max err	Residual	Gate
0%	u_{xx}, uu_x	u_{xx}: $+0.0800$, uu_x: -1.0000	0.01%	2.2×10^{-5}	PASS
1%	u_{xx}, uu_x	u_{xx}: $+0.0800$, uu_x: -1.0000	0.00%	1.9×10^{-3}	PASS
2%	u_{xx}, uu_x	u_{xx}: $+0.0800$, uu_x: -0.9999	0.02%	2.9×10^{-3}	PASS
5%	u_{xx}, uu_x	u_{xx}: $+0.0801$, uu_x: -0.9996	0.14%	8.3×10^{-3}	PASS
10%	u_{xx}, uu_x	u_{xx}: $+0.0799$, uu_x: -0.9988	0.12%	1.5×10^{-2}	PASS

KdV

σ	Support	Coefficients	Max err	Residual	Gate
0%	u_{xxx}, uu_x	u_{xxx}: -0.0484, uu_x: -1.0000	0.00%	8.3×10^{-5}	PASS
1%	u_{xxx}, uu_x	u_{xxx}: -0.0484, uu_x: -1.0001	0.03%	3.0×10^{-3}	PASS
2%	u_{xxx}, uu_x	u_{xxx}: -0.0484, uu_x: -0.9990	0.10%	6.2×10^{-3}	PASS
5%	u_{xxx}, uu_x	u_{xxx}: -0.0485, uu_x: -1.0004	0.16%	1.4×10^{-2}	PASS
10%	u_{xxx}, uu_x	u_{xxx}: -0.0485, uu_x: -1.0031	0.31%	2.6×10^{-2}	PASS

References

[1] S. L. Brunton, J. L. Proctor, J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *PNAS* 113(15):3932–3937, 2016.

[2] S. H. Rudy, S. L. Brunton, J. L. Proctor, J. N. Kutz, "Data-driven discovery of partial differential equations," *Science Advances* 3(4):e1602614, 2017.

[3] D. A. Messenger, D. M. Bortz, "Weak SINDy: Galerkin-based data-driven model selection," *Multiscale Model. Simul.* 19(3):1474–1497, 2021.

[4] D. A. Messenger, D. M. Bortz, "Weak SINDy for partial differential equations," *J. Comput. Phys.* 443:110525, 2021.

[5] H. Schaeffer, "Learning partial differential equations via data discovery and sparse optimization," *Proc. R. Soc. A* 473:20160446, 2017.

[6] P. A. K. Reinbold, D. R. Gurevich, R. O. Grigoriev, "Using noisy or incomplete data to discover models of spatiotemporal dynamics," *Phys. Rev. E* 101:010203(R), 2020.

[7] B. M. de Silva, K. Champion, M. Quade, J.-C. Loiseau, J. N. Kutz, S. L. Brunton, "PySINDy: a Python package for the sparse identification of nonlinear dynamical systems from data," *J. Open Source Softw.* 5(49):2104, 2020.

[8] A. A. Kaptanoglu *et al.*, "PySINDy: a comprehensive Python package for robust sparse system identification," *J. Open Source Softw.* 7(69):3994, 2022.

[9] N. Meinshausen, P. Bühlmann, "Stability selection," *J. R. Stat. Soc. B* 72(4):417–473, 2010.

[10] P. Zheng, T. Askham, S. L. Brunton, J. N. Kutz, A. Y. Aravkin, "A unified framework for sparse relaxed regularized regression: SR3," *IEEE Access* 7:1404–1423, 2019.

[11] G. Schwarz, "Estimating the dimension of a model," *Ann. Statist.* 6(2):461–464, 1978.

[12] Y. Kuramoto, "Diffusion-induced chaos in reaction systems," *Prog. Theor. Phys. Suppl.* 64:346–367, 1978.

[13] G. I. Sivashinsky, "Nonlinear analysis of hydrodynamic instability in laminar flames — I. Derivation of basic equations," *Acta Astronautica* 4(11–12):1177–1206, 1977.

[14] N. J. Zabusky, M. D. Kruskal, "Interaction of 'solitons' in a collisionless plasma and the recurrence of initial states," *Phys. Rev. Lett.* 15:240–243, 1965.

[15] A.-K. Kassam, L. N. Trefethen, "Fourth-order time-stepping for stiff PDEs," *SIAM J. Sci. Comput.* 26(4):1214–1233, 2005.